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THE YENEPOYA INSTITUTE OF ARTS SCIENCE COMMERCE AND MANAGEMENT

PROJECT-LOW LEVEL DESIGN

On

Identity Security Framework For E-Commerce

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1. INTRODUCTION

1.1 Purpose and Scope

Purpose:

The purpose of this Low-Level Design (LLD) document is to provide a detailed technical design of the Identity Security Framework developed for an E-Commerce platform. This document describes system modules, internal workflows, database structure, authentication mechanisms, and security implementations.

The system focuses on ensuring secure user identity management, authentication, and access control within an E-Commerce environment.

Scope:

The project includes:

- Secure user registration and login system
- OTP-based authentication via email
- Role-Based Access Control (RBAC)
- Session management
- Activity logging
- Secure product and image handling system

In Scope:

- User authentication (Login/Signup)
- OTP verification system
- Role management (Admin, Seller, User)
- Secure dashboards for each role
- Product management (Add/View Products)
- Image upload system
- Activity logging and monitoring
- Input validation and session handling

Out of Scope

- Payment gateway integration
- AI-based product recommendations
- Delivery tracking system
- Third-party integrations

1.2 Project Executive Summary

The rapid growth of online shopping platforms has increased the need for strong identity security mechanisms.

This project implements a secure identity framework that:

- Ensures only authorized users access the system
- Protects user credentials using password hashing
- Implements OTP verification for enhanced login security
- Provides role-based access for Admin, Seller, and User
- Allows sellers to securely manage products
- Tracks user activities for auditing and monitoring

The system is implemented using:

- Frontend: HTML, CSS
- Backend: Flask (Python)
- Database: SQLite

This framework improves trust, security, and controlled access in E-Commerce systems.

1.3 Constraints

- Limited to basic e-commerce features (no payment system)
- OTP depends on email SMTP configuration
- SQLite database limits scalability
- Image upload depends on correct static folder setup
- Session-based authentication is used

1.4 Document Organization

The document is structured into:

1. Introduction
2. System Overview
3. Module Design
4. Data Flow Design
5. Database Design`
6. User Interface Design
7. Security Controls
8. Testing and Validation

2. System Overview

The system follows a client-server architecture:

- User interacts through web interface
- Flask handles backend processing
- SQLite stores all system data
- Email service is used for OTP delivery

3. Module Design

3.1. Authentication Module

- Handles user registration and login
- Stores hashed passwords securely
- Validates user credentials

3.2. OTP Verification Module

- Generates OTP for login and registration
- Sends OTP via email
- Verifies OTP before granting access

3.3. Authorization Module (RBAC)

- Assigns roles: Admin, Seller, User
- Controls access to different dashboards
- Restricts unauthorized access

3.4. Dashboard Module

- Admin → Manage users and view system data
- Seller → Add and manage products
- User → Browse and view products

5. Product Management Module

- Sellers can add products
- Upload product images
- Display products dynamically

6. Activity Log Module

- Tracks user login and actions
- Stores logs for monitoring
- Admin can view system activity

4. Data Flow Design

Flow:

User enters login details

Flask validates credentials

OTP is generated and sent to email

User enters OTP

System verifies OTP

Session is created

User is redirected to respective dashboard

Activity is logged

5. Database Design

Users Table

- id
- user_id
- email
- password (hashed)
- role

Products Table

- id
- name
- price
- image
- seller_id

OTP Table

- id
- user_id
- otp
- expiry_time

Activity Logs Table

- id
- user_id
- action
- timestamp

6. User Interface Design

- Login Page
- Registration Page
- OTP Verification Page
- User Dashboard
- Seller Dashboard
- Admin Dashboard

UI is designed using HTML and CSS with a modern dark theme, centered layout, and responsive design.

7. Security Controls

- Password hashing using secure algorithms
- OTP-based authentication
- Role-Based Access Control (RBAC)
- Session management
- Input validation
- Protection against SQL Injection
- Secure file upload handling

8. Testing and Validation

- Unit testing for authentication and OTP
- Functional testing of all modules
- Security testing for vulnerabilities
- Manual testing for UI and usability

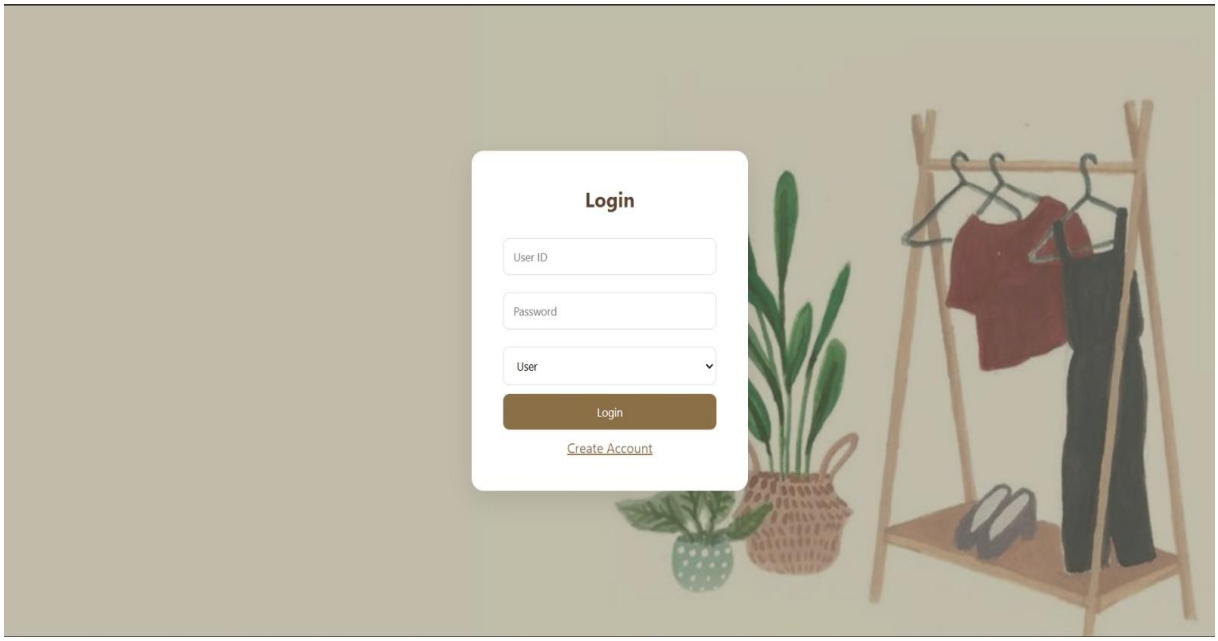
1.6 Project References

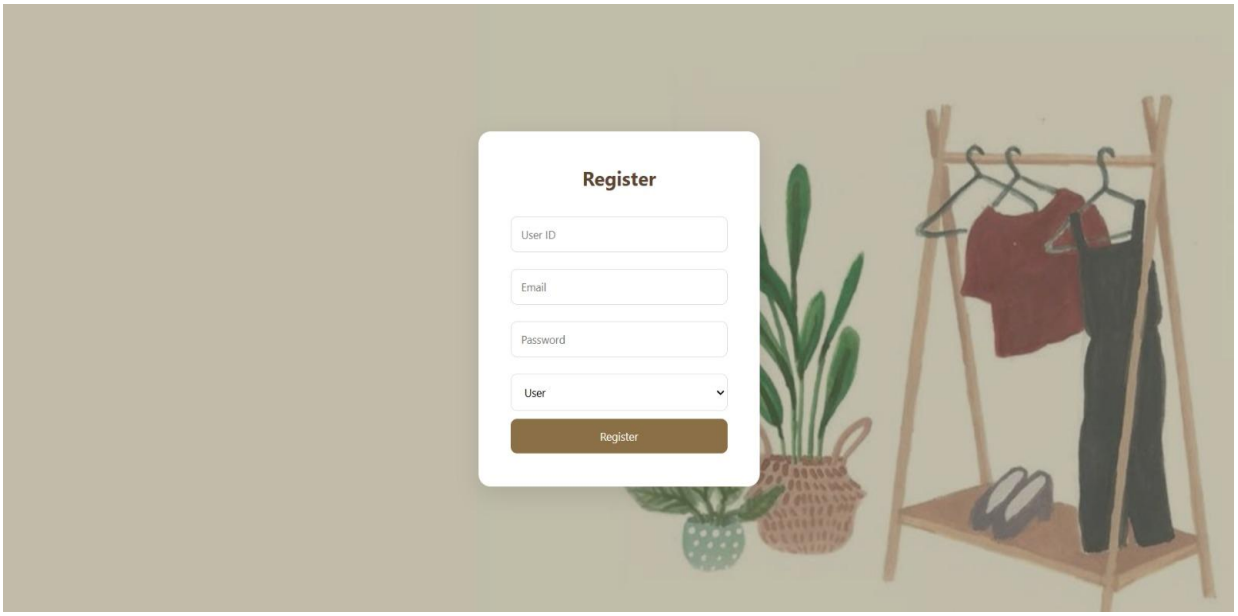
- Flask Documentation
- SQLite Documentation
- OWASP Security Guidelines
- figma

1.7 Glossary

- RBAC: Role-Based Access Control
- OTP: One-Time Password
- Flask: Python web framework
- SQL Injection: Malicious SQL attack
- Session: Temporary user login state

FRONT-END


Login
User ID
Password
User
Login
Create Account


Register
User ID
Email
Password
User
Register

BACK-END – FLASK

app.py > send_otp

```python
from flask import Flask, render_template, request, redirect, session
import sqlite3, random, smtplib, os
from werkzeug.security import generate_password_hash, check_password_hash
from email.mime.text import MIMEText

app = Flask(__name__)
app.secret_key = "secretkey"

UPLOAD_FOLDER = 'static/uploads'
app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER


if not os.path.exists(UPLOAD_FOLDER):
    os.makedirs(UPLOAD_FOLDER)

# ---------------- EMAIL CONFIG ----------------
EMAIL = "identityecommerce@gmail.com"
APP_PASSWORD = "cisugujynvwnoxfz"

def send_otp(receiver_email, otp):
    try:
        msg = MIMEText(f"Your OTP is: {otp}")
        msg['Subject'] = "OTP Verification"
        msg['From'] = EMAIL
        msg['To'] = receiver_email

        server = smtplib.SMTP_SSL('smtp.gmail.com', 465)
        server.login(EMAIL, APP_PASSWORD)
        server.send_message(msg)
        server.quit()

        print("OTP SENT:", otp)

    except Exception as e:
        print("EMAIL ERROR:", e)

# ---------------- DATABASE ----------------
def get_db():
    conn = sqlite3.connect("database.db")
    conn.row_factory = sqlite3.Row
    return conn

def init_db():
    conn = get_db()
    cursor = conn.cursor()
```